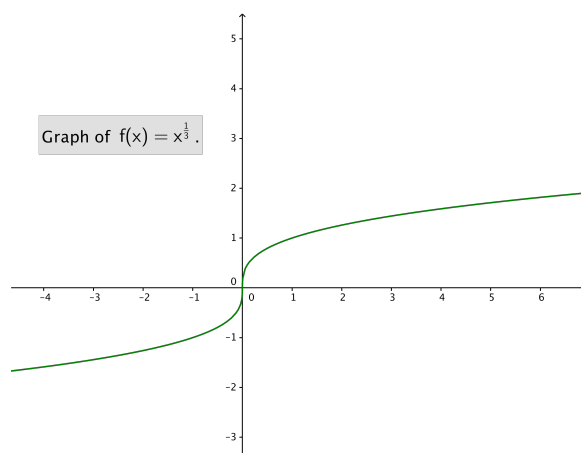


Problem 4

Problem 4

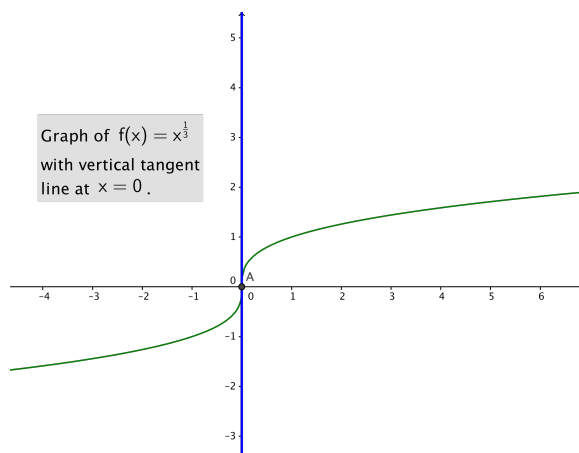
Problem 1 Define the function f by $f(x) = x^{1/3}$ and consider the graph of this function:



Which of the following two statements are true:

- (a) The graph of f has a tangent line at $x = 0$.

Explanation. This statement is **true**! The function f has a vertical tangent at $x = 0$:



(b) *The derivative $f'(0)$ is defined.*

Explanation. This statement is **false!**

$$\begin{aligned} \lim_{h \rightarrow 0^+} \frac{f(0+h) - f(0)}{h} &= \lim_{h \rightarrow 0^+} \frac{h^{1/3}}{h} \text{ which has form } \frac{0}{0} \\ &= \lim_{h \rightarrow 0^+} \frac{1}{h^{2/3}} \\ &= \infty \text{ because the limit is of the form: } \frac{\text{pos}}{0^+} \\ &\implies f'(0) \text{ is undefined} \end{aligned}$$